

From Wire to Flight: A Cross-Layer Benchmark and a Composed Network-to-Navigation Guarantee for Autonomous UAVs

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Abstract—Security research for unmanned aerial vehicles (UAVs) is split down the middle. Network intrusion detection catches malicious traffic on the swarm mesh, the command link, and the on-board bus, but says nothing about the flight; certified robustness bounds a navigation model’s error but assumes the perturbation arrives from nowhere. Neither answers the question an operator actually asks: *when an attack slips past the detector, does the aircraft still complete its mission?* We close the gap from both ends. First, we build the end-to-end benchmark that makes the question measurable: six datasets spanning the network and autonomy layers, unified under a schema whose central object is a *cross-layer pairing*—a network attack window joined to the navigation degradation it induces, tagged with an auditable flag recording whether that coupling was measured on one platform, physically modelled, or aligned in simulation. Second, we prove on exactly those objects a *composition theorem*: for any attack evading a detector at operating point θ , the navigation stack retains a certified Mission-Completion-Rate $\text{MCR} \geq f(\delta(\theta))$. Two measurements make the guarantee sharp rather than nominal. The residual attack budget *saturates* at the network’s contact-window slack—verified as at most 4.84 s of undetected delay over 15,392 observed hops—and the delay-to-position interface, calibrated on real GPS-spoofing and jamming flights on the target autopilot family, is an order of magnitude tighter than the kinematic worst case. Under that measured interface the detector’s operating point falls *inside* a nonempty certified window, and the composed guarantee is the only configuration with a nonzero certified floor against an evading attacker, dominating both single-layer baselines at every operating point (Wilcoxon, Holm-corrected, $p < 10^{-20}$). We release the schema, six validated adapters, the harness, and the certificate engine as an open artifact in which every reported number carries a provenance tag.

I. INTRODUCTION

A modern autonomous UAV is a stack of trust boundaries. Commands descend from a ground station over a command-and-control (C2) link; peer aircraft exchange state over a multi-hop swarm mesh; motor controllers and sensors talk over an on-board bus; and a navigation pipeline fuses satellite, inertial, and visual measurements into the state estimate that closes the flight control loop. An adversary can enter at any of these boundaries, and the security literature has produced strong, specialised defences at each. What it has *not* produced is a way to reason across them.

This gap is not academic. Electronic-warfare environments defeat drones not by breaking cryptography but by degrading the physics: a jammer raises the jamming-to-signal ratio (J/S) until the satellite-navigation solution collapses and the mission fails [1]. A forwarding attack in the mesh does not exfiltrate data; it delays a correction until the aircraft is acting on stale state [2], [3]. In every case the *network* event and the *navigation* outcome are two ends of one causal chain—and yet neither the datasets nor the guarantees represent that chain.

Consider what a reviewer of a UAV intrusion detector can and cannot check today. They can verify that the detector flags a delayed forwarding event or a spoofed navigation message with high recall. They *cannot* verify the claim that matters operationally—that a detector tuned to a given false-alarm budget keeps aircraft completing their missions—because no dataset records both the network event and the flight it degrades. Symmetrically, ask a certified-robustness researcher whether the UAV is safe and the answer is “the navigation model is provably stable for input perturbations up to radius R ”: a statement about a model, not a flight, and one that never says where a radius- R perturbation comes from or how often an attacker can produce it.

We close the gap from both ends, and the two halves are inseparable: the guarantee needs objects the benchmark defines, and the benchmark needs the guarantee to show what those objects are for.

The benchmark half. We unify six datasets—four established network-layer corpora and two navigation-layer sources—under a two-layer schema whose unit of analysis is neither a packet nor a flight but a *cross-layer pairing*: a network attack window W_n joined to an autonomy degradation window W_a , carrying the detector operating point, the residual undetected budget, the induced perturbation, and the mission outcome. Because a cross-layer corpus invites exactly one dangerous mistake—asserting that a network event caused a flight degradation the two were never observed together—every pairing carries a mandatory `pairing_basis` flag recording how the coupling was established.

The guarantee half. On those objects we prove a chained

statement,

$$\theta \mapsto \Delta(\theta) \mapsto \delta(\theta) \mapsto \text{MCR} \geq f(\delta(\theta)), \quad (1)$$

read left to right: a detector operating point θ bounds the residual undetected attack $\Delta(\theta)$; the residual attack maps to a bounded perturbation $\delta(\theta)$ on the navigation input; and a Lipschitz–Grönwall trajectory-tube argument turns that bounded input into a guaranteed Mission-Completion-Rate (MCR) floor. Each arrow is individually grounded in prior work—detector thresholds, worst-case attack analysis, certified robustness—but the *composition* is new: no prior guarantee chains a network-security threshold to a navigation-safety floor.

Contributions

- **A two-layer schema and six validated adapters (§V, §VI).** A single representation for any observation from any of the six sources, with lossless ingestion and a per-record honesty flag that makes the strength of every cross-layer coupling explicit and auditable.
- **A detector-operating-curve harness (§VII).** We make the network detector’s operating point a swept parameter and measure recall and false-positive rate as a function of it, producing the network-side premise a composed guarantee consumes. Across three topologies the curve has a universal knee-and-floor shape whose floor is the attacker’s self-exposure probability.
- **An empirically tightened interface (§VIII).** We prove a residual-budget lemma and then *tighten it by measurement*: the undetected budget saturates at the contact-window slack, verified at ≤ 4.84 s over 15,392 hops. We calibrate the delay-to-position rate on three real spoofing and jamming flights, obtaining a bound an order of magnitude tighter than the kinematic worst case.
- **A composition theorem and a four-certificate architecture (§IX, §X).** We prove (1) and its soundness, and show precisely why the deterministic core does not subsume randomized smoothing, PAC-Bayes, or multiplicative-weights regret.
- **Instantiation, dominance, and an open artifact (§XI).** The composed guarantee is the only configuration with a nonzero certified floor for an evading attacker; under the measured interface the detector’s operating point lies inside the certified window. Every figure and table in this paper is generated from committed result files, and every number carries a provenance tag.

Organization. §II gives the background a reader outside UAV security needs, with tables of every acronym and symbol. §III states the threat model and formalises the missing object. §IV positions the work. §V–§VII build the benchmark; §VIII–§X build the guarantee; §XI instantiates and evaluates it. §XII–§XVI discuss, bound, and conclude.

II. BACKGROUND AND TERMINOLOGY

This section is written for the reader who reviews security or machine-learning work but does not specialise in drones.

Table I collects the acronyms and Table II the mathematical symbols; both are the single reference for the rest of the paper.

A. How an autonomous UAV stays on course

An unmanned aerial vehicle (UAV, colloquially a drone) closes a control loop around a *state estimate*. An onboard computer fuses fixes from a Global Navigation Satellite System (GNSS, of which the American GPS is one constellation) receiver, inertial measurements from accelerometers and gyroscopes (the IMU), and barometric height into a best guess of where the aircraft is and how fast it is moving. On the PX4 autopilot family used throughout this paper the fusion is performed by an extended Kalman filter (EKF)—a standard recursive estimator that weighs each sensor by its expected noise [4]. The flight controller steers to keep the estimate on the planned route.

Two properties of this loop drive everything that follows. First, the loop is only as good as its *freshest trustworthy input*: if GNSS is denied, the EKF “coasts” on inertial data and its position error grows with time. Second, the loop is *networked*. Swarm members and the ground control station exchange corrections and commands over radio links: a C2 link (typically the MAVLink protocol), a multi-hop mesh in which peers relay each other’s traffic during periodic contact windows—a delay-tolerant network (DTN)—and an intra-vehicle bus (UAVCAN/DroneCAN) connecting the flight controller to motor controllers and sensors. An attacker on the mesh can therefore reach the navigation loop without ever touching the radio-frequency (RF) front end.

B. The two attacker technologies

Electronic warfare (EW) attacks the sensing physics. A *jammer* raises J/S until the receiver loses satellite lock; a *spoofers* broadcasts counterfeit satellite signals so the receiver locks onto a false position [5], [6]. *Network attacks* target the messages: a *time-delay attack*—the class our theorem covers—has a compromised relay hold packets before forwarding them, so downstream consumers act on stale state [2], [3]. Neither needs to break cryptography; both degrade the control loop through legitimate-looking inputs, which is precisely why detection and robustness must be composed rather than assumed.

C. What a “certificate” promises

A *certified* robustness statement differs from an empirical evaluation: instead of reporting that attacks in a test set were survived, it proves that *every* perturbation within an explicitly stated budget is survived. The price is conservatism—certified bounds are worst-case, so a certified floor typically sits below what is observed empirically. We keep the two kinds of number separate and label them throughout: *certified* floors come from proofs, *empirical* rates from measured flights. We use two flavours of certificate: deterministic Lipschitz-type bounds (if a model cannot amplify input changes by more than a factor L , a bounded input change yields a bounded output change) and probabilistic randomized-smoothing bounds (adding calibrated

TABLE I: Acronyms, in order of first use.

UAV	unmanned aerial vehicle (drone)
C2	command and control (ground-to-aircraft link)
MAVLink	Micro Air Vehicle Link, standard UAV C2 protocol
GNSS / GPS	Global Navigation Satellite System / Global Positioning System
IMU	inertial measurement unit
EKF	extended Kalman filter (onboard sensor-fusion estimator)
DTN	delay-tolerant network (store-and-forward mesh)
TWiG	time-window graph (the forwarding-schedule model)
UAVCAN	UAV Controller Area Network (intra-vehicle bus)
RF / EW	radio frequency / electronic warfare
J/S	jamming-to-signal power ratio, in dB
IDS / FPR	intrusion detection system / false-positive rate
MCR	Mission Completion Rate
DO-326A	RTCA/EUROCAE airworthiness-security process standard
CAF	cross-ambiguity function (receiver-level signal feature)
MWU / KL	multiplicative-weights update / Kullback–Leibler divergence
RTA	runtime assurance (verified fallback architectures)
NS-3	discrete-event network simulator

noise and taking a majority vote yields a radius within which the decision provably cannot flip) [7]. The regulatory anchor for “mission survived” is the airworthiness-security process standard DO-326A/ED-202A [8], whose industry-acceptable completion floor of 0.90 appears in our figures as a horizontal reference line.

III. THREAT MODEL AND PROBLEM FORMULATION

A. The two attack surfaces

We partition the UAV attack surface into two *layers*. The **network** layer covers everything between nodes: the swarm mesh (multi-hop routing among aircraft), the C2 link, and the intra-UAV bus. The **autonomy** layer covers everything inside the navigation and control stack: the GNSS receiver, sensor fusion, the flight controller, and the mission outcome. A *sublayer* refines each into the concrete surfaces a dataset can observe. Fig. 1 places the six datasets on the causal chain.

B. Adversary and detector

We assume an adversary who can (i) inject, drop, delay, or forge traffic on one or more network sublayers, and (ii) manipulate the RF environment of the GNSS receiver. We assume a network-layer *detector* D_θ operating at a tunable point θ that trades detection recall against false-positive rate: for the multi-hop forwarding detector we build on, θ is a per-hop deviation tolerance in seconds, and a hop is flagged if its forwarding delay exceeds θ or its next hop deviates from a scheduling oracle. Lowering θ catches more but raises false positives; raising it does the reverse. An *evading* attacker chooses its behaviour to stay just under θ and is therefore invisible by construction. We assume the adversary knows θ ;

TABLE II: Mathematical notation. Every symbol is defined where first used; this table is the single reference.

W_n, W_a	network / autonomy attack window
P	cross-layer pairing tuple, (2)
$x(t) \in \mathbb{R}^n$	navigation state (position, velocity, ...) at time t
$u(t), \delta u$	sensor input to the navigation dynamics; its perturbation
F	drift (right-hand side) of the navigation dynamics (6)
T	mission horizon (normalised to 1 in the instantiation)
m	safe-corridor margin: allowed lateral deviation, metres
$D_\theta, \theta (\epsilon)$	network detector; its operating point (tolerated per-hop residual delay, s)
s, P_c	contact-window slack (5 s); contact period (60 s)
H	number of attacker-controlled (malicious) hops on a route
$\Delta(\theta)$	residual budget: worst undetected cumulative delay, s
$\delta(\theta)$	induced perturbation on the navigation input
γ	delay-to-position rate: metres of error per second of degradation
$v_{\max}$	kinematic worst-case speed bound (15 m/s)
$L, \hat{L}, \hat{L}_{\text{loc}}$	(estimated / local) Lipschitz constant of F
$\rho(\theta)$	trajectory-tube radius $\delta(\theta)e^{LT}$, metres
$\theta^*(m)$	loosest θ whose tube fits margin m (window edge)
$f(\cdot)$	certified MCR floor as a function of the perturbation budget
R, σ, α, n	certified input radius; smoothing noise, confidence, samples
$R_{\text{emp}}, \text{KL}$	empirical risk and prior-posterior divergence, (11)
$ S $	size of the defender’s policy set in the MWU certificate (4)

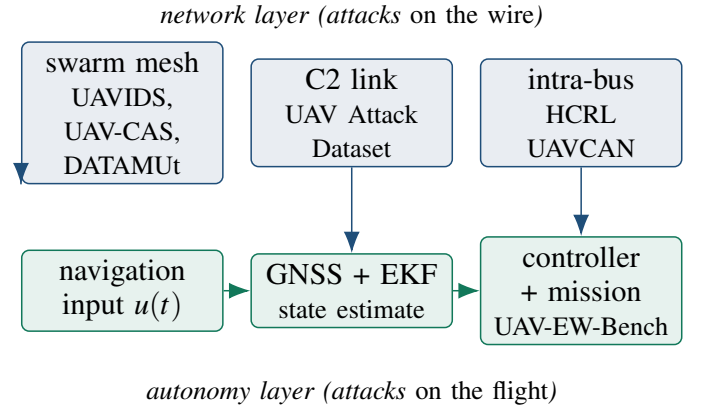


Fig. 1: The attack surfaces the six datasets cover, arranged on the causal chain. Network-layer events (top) propagate downward into the autonomy layer (bottom) through the navigation input; representing that downward arrow explicitly, rather than assuming it, is what the benchmark contributes and what the theorem then exploits.

this is the strongest reasonable assumption and the one our worst-case analysis requires.

C. Mission completion and the regulatory floor

We adopt *Mission Completion Rate* (MCR)—the fraction of flights that reach their objective under attack—as the operational metric, and take $\text{MCR} \geq 0.90$ as an operational

reference consistent with the airworthiness-security posture of DO-326A/ED-202A [8]. Emerging AI-in-aviation assurance guidance [9] motivates treating a certified MCR floor as a first-class safety property, which is what §IX delivers.

D. The missing object

Let a network attack occupy a time window $W_n = [t_0, t_1]$ on some network sublayer, and let the induced navigation degradation occupy a window W_a on the autonomy layer. Existing datasets provide either W_n (with packet labels) or W_a (with flight outcomes) but never the *pair* together with the mechanism linking them. We formalise the missing object as a tuple

$$P = (W_n, W_a, \theta, \Delta(\theta), \delta, \text{MCR}), \quad (2)$$

where θ is the detector operating point, $\Delta(\theta)$ the worst-case residual effect of an attacker that evades D_θ , δ the resulting perturbation budget on the navigation input, and MCR the measured or certified outcome. Representing P —not just W_n or W_a —is the benchmark’s contribution; bounding the last component of P from the first is the theorem’s. Crucially, the honesty of P depends on *how* W_n and W_a were linked, which we record explicitly (§V).

Definition 1 (Composed guarantee). *Given a detector D_θ and navigation dynamics (6), a composed guarantee is a function f such that for every attack a with $D_\theta(a) = \text{“clean”}$, the mission completes with rate $\text{MCR}(a) \geq f(\delta(\theta))$, where $\delta(\theta)$ bounds the navigation-input perturbation any such a can induce.*

IV. RELATED WORK

We organise prior work along the two halves of the paper: what exists on the data side (corpora and their construction), and what exists on the guarantee side (bounding a model, a controller, or a mission). The recurring gap is that each community works within one layer and leaves the interface to assumption.

a) UAV IDS datasets and surveys: Recent taxonomies of AI-enhanced UAV intrusion detection [10] document a proliferation of datasets and a recurring weakness: general-purpose network corpora lack UAV-specific protocol semantics, and UAV-specific corpora each cover a single sublayer. Notably, models trained on generic benchmarks misclassify a large fraction of GPS-spoofing events on real autopilot logs [10]—direct evidence that the network and navigation layers cannot be treated as one undifferentiated feature space. FANET security surveys [11], [12] catalogue routing attacks but stop at network effect. On the swarm mesh, UAVIDS-2025 [13], [14] provides 122,171 NS-3 flow records across four attack classes; on the C2 link, the UAV ATTACK DATASET [15], [16] records real PX4 flights under GPS spoofing/jamming and MAVLink ping-DoS; on the intra-UAV bus, the HCRL UAVCAN dataset [17] captures flooding, fuzzing, and replay. Each is authoritative for its sublayer; none reaches the flight.

b) Why dataset construction is itself a contribution: The intrusion-detection community has learned that a benchmark’s *construction* determines the validity of everything built on it. Audits of the widely used CIC-IDS2017 corpus uncovered labelling errors, flawed flow construction, and unrealistic temporal separation that materially inflate reported performance [18], [19]. A cross-layer benchmark adds a new failure mode: asserting a causal link between a network event and a flight degradation that were never observed together. Our mandatory per-record honesty flag (§V) exists precisely to bound that risk, making provenance discipline—not merely scale—a design requirement.

c) Positioning against the closest prior art: The nearest neighbour in ambition is UAV-CAS, a large AERPAW-calibrated swarm dataset that is the first to include *collaborative* multi-attack compositions and to calibrate a simulator against real testbed measurements [20]. It is an excellent network-layer benchmark and we ingest it as one of our six sources; but it remains within the network layer, and its authors explicitly list GNSS spoofing and jamming as future work. Our contribution is complementary: rather than deepen one layer we *span* both and make the coupling the unit of analysis. Where UAV-CAS asks how well a collaborative mesh attack can be detected, we ask what that attack, detected or missed, does to the flight.

d) Certified robustness and its lift to control: Randomized smoothing [7], [21] gives a probabilistic ℓ_2 certified radius that scales to large models; Lipschitz-based certification [22], [23] gives deterministic bounds that are known to be loose globally. Neural ODEs [24] admit Grönwall-type robustness arguments [25], the continuous-time analogue we use. These bound a *model’s* output. Lifting to sequential decision-making, CROP certifies a cumulative-reward lower bound for policies under state perturbations [26] and CARRL certifies robust actions [27]; reachability tools bound trajectories of network-controlled systems [28], [29]; control-theoretic work provides safety under sensor false-data injection [30]. CROP is the closest analogue to a certified mission floor, but it certifies a policy against a *given* perturbation ball—it does not connect that ball to a network-security operating point.

e) Secure state estimation, runtime assurance, delay attacks: A parallel control-theoretic line asks how much sensor corruption an estimator can provably tolerate: exact recovery is possible below a sensor-corruption threshold [31], the combinatorial search is practical with SMT solvers [32], and recent surveys catalogue the field [33]. These bound estimation error given *which* sensors are compromised; they do not model a detector whose threshold shapes what the attacker can inject in the first place—the premise our Lemma 1 supplies. The Simplex architecture and its descendants [34]–[36] guarantee safety by switching from an unverified controller to a verified baseline when a monitor fires; our certified window is exactly the region in which the high-performance policy may run, and we supply the quantitative boundary such architectures otherwise hand-tune. Finally, the effect of injected delay on

feedback stability is well studied, with detectors based on state-aware inference or divergence tests [2], [3], [37]; these quantify *whether* a delay attack is dangerous or detectable, but none carries the surviving delay through to a certified mission floor.

f) *Composing guarantees—the crux:* The closest prior art composes certificates *within* an ML pipeline. [38] jointly certifies a classifier with a detection class, so an input is provably flagged or correctly classified—structurally our evade-or-bound dichotomy, but single-layer. [39] decomposes end-to-end certified robustness of a two-stage sensing→reasoning pipeline, the strongest template for chained certification, but stays within perception. Assume-guarantee and contract-based CPS design [40] give the compositional vocabulary but not ML-robustness certificates. None chains a *network-intrusion-detection* guarantee to a *navigation* certificate. That is our gap, and closing it requires the benchmark and the theorem together.

V. THE TWO-LAYER SCHEMA

The schema has three record kinds, designed so the cross-layer pairing is a first-class object rather than an implicit join.

Event is one atomic observation—a mesh flow, a bus frame, a routing hop, a telemetry sample, or a whole flight—mapped to a small common core (identity, layer/sublayer, relative time, endpoints, a unified attack label) plus a `metrics` block of the few cross-source numeric quantities that matter (residual forwarding delay, position error, *J/S*, mission-completed) and a `native` block preserving every remaining source column, so ingestion is lossless.

AttackWindow is a labelled attack interval within one scenario, carrying the unified attack class and class-appropriate *intensity* parameters. It is the unit of cross-layer alignment.

CrossLayerPairing realises the tuple P of (2): it joins a network window to an autonomy window and records θ , the residual budget $\Delta(\theta)$, the perturbation δ with a *named, versioned* mapping, the certificate hook, and the empirical MCR. Its `pairing_basis` field is the schema’s conscience: `measured_same_platform` (both windows recorded on one testbed—the strongest evidence), `physics_model` (coupled through a stated physical model), or `synthetic_alignment` (aligned by construction). A reviewer can filter the benchmark by evidence strength; a claim can never silently upgrade its own provenance.

Three design choices let six heterogeneous sources coexist without information loss. Time is always *relative* to each source’s own capture start, because the sources share no global clock; cross-layer alignment happens at the window level, so a mismatch between a 50 Hz telemetry stream and a per-flow network record never forces a lossy resample. Attack labels are *unified* into a common taxonomy while the verbatim source label is retained in `label_native`, so a model can train on the unified class and an auditor can still recover what the source called it. And every unmapped column is preserved,

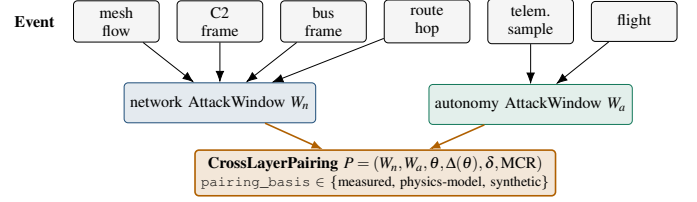


Fig. 2: The schema in one picture. Heterogeneous events from six datasets fold into typed network and autonomy *AttackWindows*; a *CrossLayerPairing* joins one of each and records the detector operating point, the residual budget, the induced perturbation, and the mission outcome, tagged by how the coupling was established.

which makes the ingestion provably lossless rather than merely convenient.

VI. THE CORPUS AND INGESTION ADAPTERS

Table III summarises the six datasets and the layer each anchors; Table IV reports what was actually ingested. Every adapter emits schema-valid records, and unmapped source columns are preserved. We do not re-host raw third-party data; the artifact ships integration scripts and unified labels, and each source is fetched from its origin.

A. What each adapter does

An adapter’s job is lossless normalization. Three warrant specifics because their sources are irregular. The UAV ATTACK DATASET adapter parses PX4 flight logs, computes horizontal position error against a benign reference (or, absent one, the flight’s own pre-attack median), and emits the `measured_same_platform` pairings that ground the interface mapping of §VIII. The HCRL UAVCAN adapter parses candump-style bus frames; because the per-scenario attack labels are not machine-readable in the release, we *derive* each scenario’s class from the frames themselves—burst rate relative to the normal bus rate, the fraction of distinct payloads, and per-frame byte entropy cleanly separate a flooding burst (one payload hammered above bus rate) from a fuzzy burst (near-random payloads) from a replay burst (a small set of captured payloads reused). Table V reports the result, flagged for confirmation against the dataset’s technical report. The UAV-CAS adapter parses the calibrated digital twin’s flow statistics including its stringified list columns; it is verified against CSVs produced by the original authors’ own generator.

VII. THE DETECTOR-OPERATING-CURVE HARNESS

A cross-layer guarantee needs the detector’s behaviour *as a function of its operating point*. We expose θ as a swept parameter and, for each value, measure detection recall and false-positive rate over multiple seeds, three network topologies, four routing policies, and two attacker delay regimes.

The resulting operating curve (Fig. 3) is the network-side input a composed guarantee consumes: it tells us, for any target false-positive budget, how large an undetected per-hop delay

TABLE III: The six-dataset corpus. Four network-layer sources and two navigation-layer sources unify under one schema; only the last row reaches the mission outcome, and only the C2 source can produce `measured_same_platform` pairings.

Dataset	Real/Sim	Layer · sublayer	Event kind	Unique role in the end-to-end benchmark
UAVIDS-2025 [13]	Sim (NS-3.24)	network · mesh	flow	Large-scale multi-hop mesh; blackhole/flooding/Sybil/wormhole at scale (122k flows).
UAV-CAS [20]	Sim (digital twin)	network · mesh	flow	AERPAW-calibrated swarm; collaborative multi-attack compositions (99k flows, 1024 configs).
UAV ATTACK DATASET [15]	Real testbed	network · C2 (MAVLink)	telemetry	Real PX4 GPS spoof/jam + ping-DoS on the <i>same</i> airframe family we model—the measured bridge.
HCRL UAVCAN [17]	Real testbed	network · intra-bus	frame	On-board UAVCAN/DroneCAN flooding/fuzzy/replay—a third attack surface below the C2 link.
DATAMUT traces [2]	Sim (event)	network · mesh (DTN)	hop	Time-window-graph forwarding detector with a <i>sweepable</i> operating point θ .
UAV-EW-BENCH-2026 [41]	Sim (physics-informed)	autonomy · GNSS+ctrl	flight	The navigation anchor: full 0–40 dB J/S sweep with per-flight MCR outcomes.

TABLE IV: Real ingested corpus. Counts are from schema-validated adapter outputs; every record validates against the schema of §V.

Source (real data ingested)	Events	Windows
UAV-EW-BENCH-2026	93,600	468
UAV ATTACK DATASET (live sample)	610	0
UAVIDS-2025 (shard 0)	122,171	4
HCRL UAVCAN (10 scenarios)	200,000	34
UAV-CAS (stat)	adapter ready; file pending	
DATAMUT traces	generated per sweep (15,392 hops)	

an evading attacker may inject—the residual budget $\Delta(\theta)$ of (2). The knee in the curve is structural: an evading attacker’s per-hop delay is additive up to the inter-node contact-window slack, beyond which a missed contact window forces a full-period penalty.

The knee, and the floor it descends to, are not simulation artifacts but a structural property of any store-and-forward schedule, and they recur across all three topologies. Two observations transfer directly to the guarantee. First, the floor height differs by topology because it equals the probability that a malicious delay overshoots the slack and self-exposes—higher in the medium delay regime ($U[1,10]$ s) than the low one ($U[1,7]$ s), because a wider delay draw is likelier to overshoot. Second, no topology’s recall falls to zero: past the knee the detector still catches the self-exposing fraction, which is precisely why an evading attacker’s undetected budget *saturates* rather than growing without bound. That is the empirical basis for Lemma 1.

TABLE V: Frame-derived attack classes for the HCRL UAVCAN scenarios, inferred from burst rate, payload-distinct share, and byte entropy; scenarios exhibiting more than one burst shape map to *collaborative*.

Scenario	Frames	Bursts	Frame-derived class(es)
type1	207,858	3	flooding
type2	134,170	3	replay
type3	197,479	3	fuzzy
type4	133,374	3	fuzzy
type5	180,608	3	fuzzy
type6	241,321	4	fuzzy
type7	234,162	4	replay, fuzzy
type8	265,800	4	fuzzy
type9	230,378	4	replay, fuzzy
type10	207,380	3	replay, fuzzy

VIII. THE INTERFACE MAPPING $\theta \mapsto \delta(\theta)$

The interface is where a network-security quantity becomes a navigation-security quantity. It is the crux of the composition, and—as §XI shows—the quantity that actually decides whether a useful detector setting is certifiable.

A. Residual budget of an evading attacker

Lemma 1 (Residual delay budget). *Let D_θ flag any forwarding hop whose injected delay exceeds θ . An attacker that evades D_θ injects at most θ of undetected delay per malicious hop—but only up to the contact-window slack s : a per-hop delay exceeding s misses its forwarding window, incurs a full-period penalty P_c , and is therefore flagged at any operating point. Hence undetected delay is capped at $\min(\theta, s)$ per hop,*

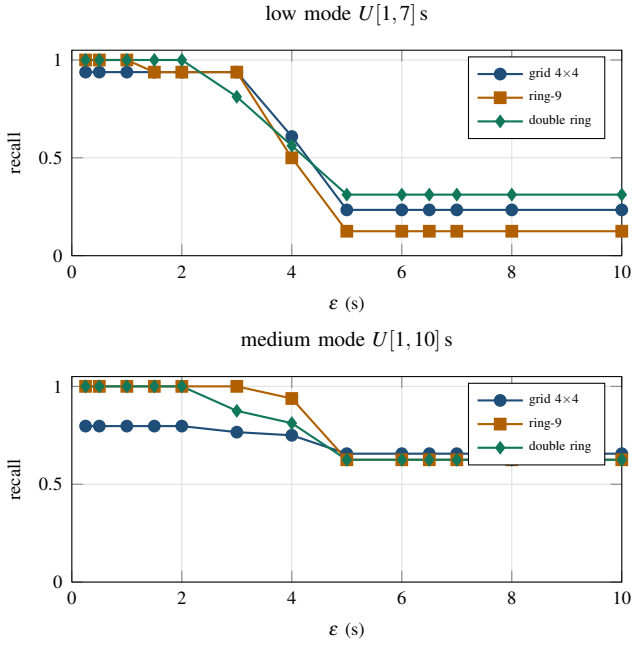


Fig. 3: The detector operating curve across all three topologies (grid, ring, double-ring), both delay regimes, 8 seeds $\times$ 4 routing policies. Recall holds near its ceiling while ϵ stays under the 5 s contact-window slack, then falls onto a floor set by the missed-window self-exposure penalty (a delay past the slack costs a full TWiG period and is flagged at any ϵ). False-positive rate is zero throughout; the measured benign residual ceiling is 0.178 s. The knee-and-floor shape is universal; the floor height is the topology-specific self-exposure probability.

and over a route with H attacker-controlled hops the worst-case undetected cumulative delay is

$$\Delta(\theta) \leq H \min(\theta, s). \quad (3)$$

Proof sketch. Evasion requires per-hop delay $\leq \theta$. If $\theta \leq s$ the delay fits inside the contact window and the additive term is $H\theta$. If $\theta > s$, any delay in $(s, \theta]$ overshoots the window, waits a full period $P_c \gg s$, and is flagged regardless of θ ; an evading attacker therefore cannot use more than s of invisible delay per hop, giving the Hs cap. Summing over hops yields (3). $\square$

The cap is the *knee*: below the slack the residual budget grows linearly in θ ; above it the budget *saturates* at Hs , so the certified floor plateaus rather than collapsing. We verify the cap directly on the released harness: over 15,392 observed hops across three topologies, two attack modes, and eight seeds, no undetected malicious per-hop residual ever exceeds 4.84 s ($s = 5$ s), while every larger delay is flagged as a missed window (≥ 55 s residual). This measurement is what turns the classic additive budget $H\theta + P_c \mathbf{1}[\theta > s]$ into the tighter, saturating (3)—an instance of the benchmark improving the theorem rather than merely testing it.

TABLE VI: The measured delay-to-position rate γ , from three real PX4 flights (one benign reference, one GPS-jamming, one GPS-spoofing) in the UAV ATTACK DATASET, self-referenced to the pre-attack median. “noise” is the pre-attack position-error 95th percentile; γ_{lsq} and γ_{peak} are the least-squares and peak-secant growth rates after attack onset. These are the measured_same_platform pairings.

Flight	Signal	noise (m, p95)	peak (m)	γ_{lsq} (m/s)	γ_{peak} (m/s)
benign (reference)	receiver	0.50	1.7	—	—
benign (reference)	ekf	0.60	1.5	—	—
GPS jamming	receiver	0.41	6.7	-0.0821	0.4805
GPS jamming	ekf	0.47	7.0	-0.1133	0.5185
GPS spoofing	receiver	1.14	14.3	0.2435	1.195
GPS spoofing	ekf	0.87	15.6	0.7007	1.365

B. From delay to input perturbation

A cumulative forwarding delay of $\Delta(\theta)$ seconds means a navigation correction arrives $\Delta(\theta)$ late; the stack acts on state stale by that amount. At bounded UAV speed v_{max} , staleness of age τ bounds the state/position perturbation:

$$\|\delta u\| \leq v_{\text{max}} \cdot \Delta(\theta). \quad (4)$$

Equation (4) is the kinematic worst case: it assumes the aircraft travels at top speed in the wrong direction for the entire staleness interval. That is sound but pessimistic, because a real estimator does not fly blind—when GNSS corrections are late or corrupted the EKF down-weights them and coasts on inertial data, so position error grows far more slowly. We therefore tighten (4) to

$$\|\delta u\| \leq \gamma \cdot \Delta(\theta), \quad \gamma \ll v_{\text{max}}, \quad (5)$$

where γ (metres of position error per second of degradation) is *measured* on real GPS-spoofing and GPS-jamming flights from the same PX4 airframe family the navigation model represents. Table VI reports the calibration: a benign reference flight fixes the pre-attack noise floor, and the jamming and spoofing flights give the growth rate after onset under two explicit choices of error signal—the raw receiver fix the attacker injects, and the EKF-filtered position the controller actually acts on. Because that choice is itself a certificate-semantics decision we make it an explicit switch and report both; the certificate consumes the EKF value, which closes the loop. The gap between the measured $\gamma \approx 1.2$ –1.4 m/s and $v_{\text{max}} = 15$ m/s is exactly the slack that decides whether the detector’s operating point is certifiable.

IX. THE COMPOSITION THEOREM

A. Navigation dynamics

Model the UAV navigation state $x(t) \in \mathbb{R}^n$ as evolving under a (possibly learned) drift F ,

$$\dot{x}(t) = F(x(t), u(t)), \quad t \in [0, T], \quad (6)$$

where $u(t)$ is the sensor/measurement input over a mission horizon T . A mission *completes* if the realised trajectory stays

within a mission-defined safe corridor for all $t \in [0, T]$; MCR is the probability of completion over the mission distribution. An attack perturbs the input, $u \mapsto u + \delta u$.

Theorem 1 (Composed network-to-navigation guarantee). *Let the navigation drift F of (6) be L -Lipschitz in its input over the operating region, with mission horizon T and safe-corridor margin m . Let D_θ be a network detector with residual budget $\Delta(\theta)$ (Lemma 1) and interface mapping $\delta(\theta)$ ((5) or the kinematic (4)). Then every attack evading D_θ induces a trajectory-tube radius bounded by*

$$\rho(\theta) = \delta(\theta) e^{LT}, \quad (7)$$

and the mission completes whenever $\rho(\theta) \leq m$; consequently

$$\text{MCR} \geq f(\delta(\theta)) = \Pr_{\text{missions}} [\delta(\theta) e^{LT} \leq m]. \quad (8)$$

Proof. Fix a mission and let $x_{\text{nom}}(t)$ be the nominal trajectory—the solution of (6) under the attack-free input $u(t)$ —and $x_{\text{att}}(t)$ the trajectory under $u(t) + \delta u(t)$, both started from $x_{\text{nom}}(0) = x_{\text{att}}(0)$. Write $e(t) = x_{\text{att}}(t) - x_{\text{nom}}(t)$. From (6), $\dot{e}(t) = F(x_{\text{att}}, u + \delta u) - F(x_{\text{nom}}, u)$, so L -Lipschitzness gives $\|\dot{e}(t)\| \leq L\|e(t)\| + L\|\delta u(t)\|$. Integrating with $e(0) = 0$,

$$\|e(t)\| \leq L \int_0^t \|e(\tau)\| d\tau + L \int_0^t \|\delta u(\tau)\| d\tau.$$

Applying the integral form of Grönwall's inequality [42] to the non-decreasing forcing term $g(t) = L \int_0^t \|\delta u\| d\tau$ yields $\|e(t)\| \leq g(t) e^{Lt}$, and bounding the input by $\|\delta u(\tau)\| \leq \delta(\theta)$ gives, at the horizon,

$$\|e(T)\| \leq \delta(\theta)(e^{LT} - 1) \leq \delta(\theta) e^{LT}, \quad (9)$$

which is (7); we report the closed-form right-hand bound. Now suppose $\rho(\theta) \leq m$. The nominal trajectory lies in the safe corridor by construction, and every point of the attacked trajectory is within $\rho(\theta) \leq m$ of it by (9); hence the attacked trajectory never exits the corridor over $[0, T]$ and the mission completes. Taking the probability of $\{\rho(\theta) \leq m\}$ over the mission distribution—the only source of randomness once θ is fixed, since $\delta(\theta)$ is a deterministic bound—gives (8). Composing $\theta \mapsto \Delta(\theta)$ (Lemma 1), $\Delta \mapsto \delta$ (§VIII), and $\delta \mapsto \text{MCR}$ establishes the chain (1). $\square$

Proposition 1 (Soundness). *The certified floor never overstates the achievable completion rate: for every θ , the true MCR of any D_θ -evading attack satisfies $\text{MCR} \geq f(\delta(\theta))$.*

Proof. Every step above is an inequality in the conservative direction: the Lipschitz bound over-estimates deviation growth, the input bound uses the worst-case $\delta(\theta)$, and the corridor test is a sufficient—not necessary—condition for completion. Hence $f(\delta(\theta))$ lower-bounds the true completion probability. The empirical validation of §XI confirms the certified floor lies below the measured rate. $\square$

Two consequences frame the rest of the paper. First, (8) is *monotone and then saturating* in θ : tightening the detector shrinks the residual budget and raises the certified floor, but

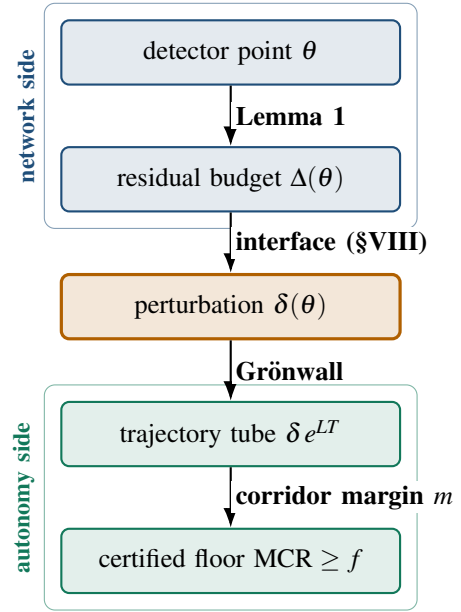


Fig. 4: The composition chain of Theorem 1. A detector operating point flows top-to-bottom through a residual-budget bound, an empirically grounded interface mapping, a Grönwall trajectory tube, and a corridor-margin test, arriving at a certified mission-completion floor. The upper two stages are network-side quantities, the lower two autonomy-side; $\delta(\theta)$ is the bridge between them.

only until θ reaches the slack s , past which the budget saturates at Hs and the floor plateaus—a quantitative statement of both how much network security “buys” in navigation safety and where that purchase stops. Second, the exponential e^{LT} in (7) is the core certificate’s Achilles heel: for large L or long horizons the deterministic bound becomes loose, motivating the complementary certificates of §X.

X. A FOUR-CERTIFICATE ARCHITECTURE

Theorem 1 uses the Lipschitz–Grönwall certificate as its spine. Does that single certificate suffice? It does not. The bound is *deterministic* and *worst-case*: it holds for any perturbation up to its radius, but the radius collapses as LT grows, and it presumes a Lipschitz constant that is tractable to bound. Three regimes escape it, each covered by a distinct certificate (Table VII).

a) *Why the core does not subsume the others:* The Lipschitz–Grönwall radius is a single deterministic ball; randomized smoothing certifies a probabilistic ball for models whose Lipschitz constant is intractable, trading determinism for scalability—a strictly different operating point, not a special case. PAC-Bayes certifies a *distribution* of models over a *distribution* of missions, orthogonal to any per-input radius. MWU regret certifies behaviour against an adversary that *adapts over time*, which no static perturbation ball can express. The four form a cover, not a hierarchy.

TABLE VII: The four certificates cover disjoint regimes; none subsumes another. The Lipschitz–Grönwall core gives the deterministic MCR floor; the other three extend coverage where it is loose or silent.

Certificate	Guarantee type	Unique niche
Lipschitz–Grönwall	deterministic, worst-case	The MCR floor: bounded input $\Rightarrow$ bounded tube $\Rightarrow$ bounded loss. Tight only for moderate LT .
Randomized smoothing [7]	probabilistic radius	Scales to large / non-Lipschitz-tractable models; certifies when the budget is known only statistically.
PAC-Bayes [43]	distributional	Transfers the floor from the training mission distribution to unseen deployment regions.
MWU regret [44], [45]	online, adaptive	Bounds loss against an <i>adaptive</i> jammer over time—a dimension no fixed-ball certificate addresses.

For concreteness we record the two complementary bounds we instantiate numerically. Randomized smoothing [7] wraps the base model in Gaussian noise of level σ and certifies, with confidence $1 - \alpha$ over n samples, an ℓ_2 radius

$$R_{RS} = \frac{\sigma}{2} (\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)), \quad (10)$$

with Φ^{-1} the inverse standard-normal CDF and $\underline{p}_A, \overline{p}_B$ confidence bounds on the top-two class probabilities. The PAC-Bayes complement [43], [46] transfers a bound from the training mission distribution to deployment: with empirical risk R_{emp} over m missions and prior-to-posterior divergence KL, with probability $1 - \delta$,

$$R_{\text{true}} \leq R_{\text{emp}} + \sqrt{\frac{\text{KL} + \ln(2\sqrt{m}/\delta)}{2m}}. \quad (11)$$

Our engine evaluates (11) against the trained model’s measured R_{emp} ; the single quantity it awaits is the numeric KL, which we flag as pending rather than substitute (Table VIII).

Which certificate governs when. The four do not compete for the same operating point; each binds in a different regime and the headline floor is their pointwise minimum. For the time-delay class this paper foregrounds, the perturbation is a bounded deterministic state staleness, so Lipschitz–Grönwall is active and sets the certified window of §XI. Randomized smoothing takes over when the budget is known only statistically—a jamming intensity characterised by a distribution rather than a hard bound. PAC-Bayes binds when a floor certified on the training mission mix must survive unseen missions. MWU regret binds over the *time* axis, when the jammer adapts across a mission. Because their binding regimes are disjoint, the minimum is never vacuous for a reason another certificate could have prevented.

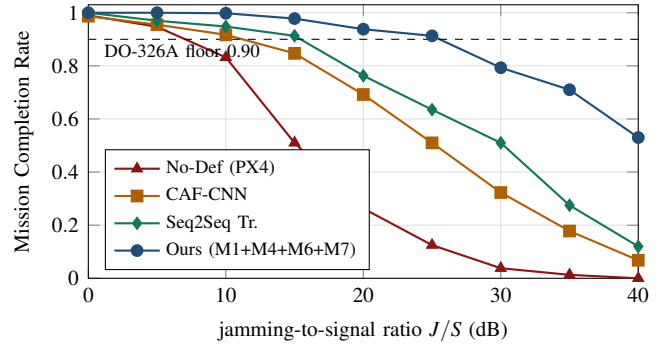


Fig. 5: The navigation anchor: MCR vs. J/S on UAV-EW-BENCH-2026, computed from the released 93,600-flight table (Wilson 95% intervals in the artifact). The graded outcome is what a network attack window is paired against.

XI. INSTANTIATION AND EVALUATION

A. The navigation anchor

The autonomy layer supplies the outcome the whole construction reaches for. Fig. 5 shows MCR versus J/S for four defence configurations, computed from the released benchmark’s 93,600-flight per-flight table. The undefended baseline collapses below the 0.90 floor by $J/S \approx 8$ dB, while the strongest configuration holds it to $J/S \approx 25$ dB. For our purpose the point is not which defence wins but that the autonomy layer produces a *graded, measurable* outcome a network event can be paired against.

B. Reference constants

We instantiate the certificate on the reference navigation model, a continuous-time graph model of the GNSS constellation and receiver. Global power-iteration gives a Lipschitz estimate $\hat{L} \approx 1.01$; measuring the constant *along the hidden-state trajectories the integrator actually visits on operating-region inputs* gives a larger local value $\hat{L}_{\text{loc}} \approx 1.18$. We report the tighter local radius as the defensible number: with $T = 1$ and $\epsilon_{\text{out}} = 0.5$, $R = \epsilon_{\text{out}} e^{-\hat{L}_{\text{loc}} T} \approx 0.153$ (the global estimate gives 0.182). Note the local radius is *smaller*: the bound is not loose in the direction that would flatter us, and the certified regime is governed by real sensitivity rather than an optimistic estimate. The smoothing complement gives a certified ℓ_2 radius ≈ 0.44 at $\sigma = 0.25$ ($\alpha = 10^{-3}$, $n = 200$; measured mean 0.446 on the trained checkpoint), and the MWU defender mixes four policies with regret $R(T) \leq \sqrt{T \ln |S|}$, $|S| = 4$. Constants come from the trained reference implementation on a transparent synthetic corpus and are held fixed; Table VIII lists each with its provenance tag.

C. A worked end-to-end example

To make (1) concrete we trace one number through it. Take the grid topology, whose routes carry $H = 2$ attacker-controlled hops, and the detector setting $\theta = 0.25$ s. (1) *Threshold to budget.* Since $\theta < s = 5$ s, Lemma 1 gives $\Delta(\theta) \leq H\theta = 0.50$ s of undetected staleness. (2) *Budget to perturbation.* With the

TABLE VIII: Instantiated constants and their provenance. The one pending numeric constant is the PAC–Bayes prior/posterior KL, which we flag rather than substitute.

Component	Constant	Provenance
Lipschitz–Grönwall	$\hat{L}_{\text{loc}}=1.181 \Rightarrow R=0.153$; $\hat{L}=1.013 \Rightarrow R=0.182$	global trained ckpt [fixture]
Rand. smoothing	$\sigma=0.25$, $R_{\ell_2}=0.44$ (measured 0.446)	trained ckpt [fixture]
PAC–Bayes	McAllester on real R_{emp} ; numeric KL pending	methods ch. [pending]
MWU regret	$R(T) \leq \sqrt{T \ln 4}$; $T=100 \Rightarrow 11.77$	exact
Unit bridge (RF classes)	caf_shift_v2 : 0.45 m (Grönwall) / 1.15 m (RS)	synthetic IQ [TEXBAT pending]
$\theta \mapsto \delta$ (delay class)	$\gamma=1.20\text{--}1.37$ m/s measured; kinematic 15 m/s fallback	real 3 flights [hover]

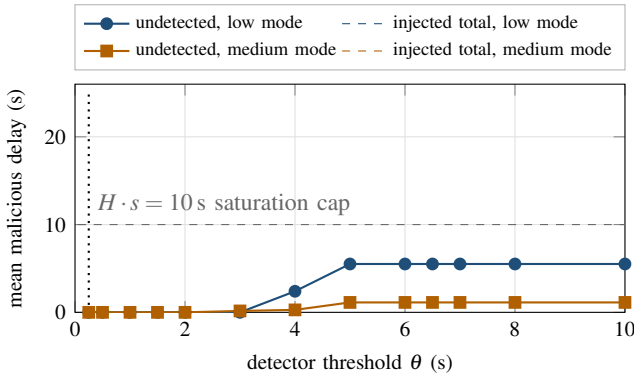


Fig. 6: The measured residual budget behind Lemma 1 (scenario 1, mean over 8 seeds $\times$ 4 policies). The *undetected* delay an evader slips past the detector (solid) saturates well below the *total* injected delay (dashed): beyond the contact-window slack the extra delay produces missed windows flagged at any θ , so the curve the certificate consumes flattens rather than growing.

measured EKF rate $\gamma = 1.37$ m/s, (5) gives $\|\delta u\| \leq 0.69$ m; the kinematic bound would instead give 7.5 m. (3) *Perturbation to tube*. With $\hat{L}_{\text{loc}} = 1.18$ and $T = 1$, the tube radius is $\rho = 0.69 e^{1.18} \approx 2.25$ m. (4) *Tube to floor*. For any corridor margin $m \geq 2.25$ m the mission is certified to complete; across the margin family $\{2, 5, 10, 20\}$ m the certified floor is positive for all but the tightest 2 m corridor. Under the kinematic bound, step (3) yields $\rho \approx 24.5$ m, exceeding every margin—which is exactly why the empirical calibration, not the theorem, decides the framing. Fig. 6 shows the measured undetected budget feeding step (2), confirming the saturation predicted by Lemma 1.

D. The certified floor and the interface that decides it

Fig. 7 is the headline result, and it is deliberately *parametric in the interface mapping*. Under the kinematic worst case, $\theta = 0.25$ s falls just outside the certified window; under the

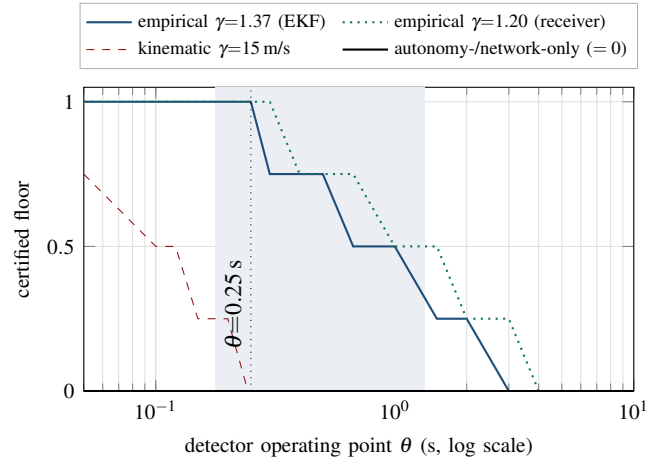


Fig. 7: The headline result: certified floor vs. detector operating point θ for the time-delay class (grid topology, Grönwall tube at the measured local $L=1.181$, $H=2$, corridor-margin family $\{2, 5, 10, 20\}$ m). Under the kinematic worst case (red) $\theta=0.25$ s is just outside the certified window; under the empirically measured mapping from real GPS-spoof/jam flights ($\gamma \approx 1.2\text{--}1.4$ m/s) it is *inside* at every margin (shaded: the EKF window $[0.178, 1.334]$ s at $m=10$ m). Both single-layer baselines certify a zero floor for an evading attacker (black).

empirically measured mapping it falls *inside* at every corridor margin. Fig. 8 quantifies how much room there is: the mapping would have to steepen past $\gamma_{\text{req}} = 7.28$ m/s (at a 10 m corridor) to push the operating point back out—more than five times the measured rate. This is the paper’s most consequential finding for the field: what determines whether cross-layer certification is useful is a *measurement* of how attacks propagate into state error, not the proof machinery.

E. Dominance over single-layer baselines

We state the dominance claim precisely rather than as blanket range-coverage. The composed guarantee dominates two baselines: (i) *autonomy-only*—a navigation certificate with no detector, which must assume the full unmitigated attack including the missed-window penalties (measured cumulative delay up to 424 s) and therefore certifies a zero floor against an evading attacker; and (ii) *network-only*—a detector with no navigation guarantee, for which any evading attack leaves the aircraft undefended, floor zero by definition. **The composed guarantee is the only configuration with a nonzero certified floor at all.** Quantifying per seed with the Wilcoxon signed-rank test and Holm correction, the composed floor exceeds the autonomy-only floor at all 13 operating points on the θ grid, every adjusted $p < 10^{-20}$. The composition wins because it alone uses the detector to *shrink* the perturbation the certificate must absorb, and—via the tightened budget of Lemma 1—the floor saturates rather than collapsing past the knee.

The honest scope: the certified operating window is nonempty but bounded. Under the empirically calibrated interface it is $[0.178, 1.33]$ s at a 10 m corridor, with the operating

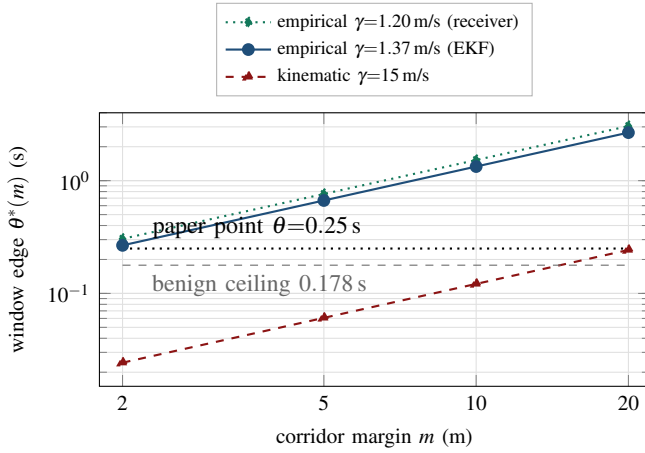


Fig. 8: The decisive sensitivity: the loosest detector setting $\theta^*(m) = m/(\gamma e^{LT} H)$ whose worst-case evading tube still fits a corridor of margin m , per interface mapping. A mapping certifies the paper operating point iff its curve lies above the $\theta=0.25$ s line; the feasible region is bounded below by the measured benign residual ceiling (0.178 s, the FPR=0 floor). The measured rates clear the requirement $\gamma_{\text{req}}=7.28$ m/s at $m=10$ m by $\sim 5\times$; the kinematic worst case fails at every margin.

point $\theta = 0.25$ s comfortably inside; under the conservative kinematic mapping it shrinks to $[0.178, 0.243]$ s and $\theta = 0.25$ s falls just outside. We claim the former and show the latter as the worst-case fallback; we do not claim coverage across the entire relaxation range.

F. Certified-versus-empirical validation, and the two floors

The certificate is sound iff the certified floor never exceeds the realised MCR (Proposition 1). At the anchor $J/S = 20$ dB the strongest defence attains empirical MCR = 0.938 over 93,600 real benchmark flights, comfortably above the certified target 0.80; the gap is the conservativeness of the worst-case Grönwall bound, and it narrows where the empirically tightened interface replaces the kinematic worst case.

We keep two floors distinct throughout, because conflating them would overstate the guarantee. The reference certificate targets a *certified* MCR ≥ 0.80 at the anchor operating point; the empirical benchmark reports the DO-326A 0.90 *operational* floor holding further into the band. These are different quantities—a conservative proof-derived bound versus a measured rate—and we label every reported number accordingly.

G. Three evidence classes, one worked pairing each

Table IX reports the pairing at a representative operating point. The `pairing_basis` flag is easiest to understand through one example of each class. *Measured, same platform*: the UAV ATTACK DATASET records a real GPS-jamming flight and its measured position error on one airframe; a pairing joining that flight to the corresponding RF event is

TABLE IX: A representative cross-layer pairing: a detector-evading time-delay attack still induces measurable navigation staleness. Values from the released artifact (harness simulation; three-flight δ calibration; benchmark MCR).

Quantity	Value at representative θ
Detector threshold θ	0.25 s (paper operating point)
Detector verdict on tuned attacker	evaded (by construction)
Malicious hops H	2 (grid/ring/double-ring routes)
Residual budget $\Delta(\theta) \leq H \min(\theta, s)$	≤ 0.50 s ($s=5$ s; measured cap 4.84 s)
Measured undetected delay (192 runs)	median 0.00 s, max 0.00 s
Staleness, empirical $\gamma=1.37$ m/s	≤ 0.69 m (Whelan, hover)
Staleness, kinematic $v_{\text{max}}=15$ m/s	≤ 7.5 m (worst case)
Paired autonomy window	UAV-EW-BENCH at $J/S=20$ dB
Empirical MCR of paired window <code>pairing_basis</code>	0.938 (strongest defence) synthetic / measured (per record)

tagged `measured_same_platform`, and its δ is a measurement, not a model. *Physics model*: a forwarding-attack window carries a residual budget in seconds; coupling it to a navigation window through the stated relation (5) yields a `physics_model` pairing—a named, versioned physical relation a reviewer can inspect and challenge, and the class the certificate consumes. *Synthetic alignment*: when a network window from one simulated source and an autonomy window from another are aligned by construction, the pairing is tagged `synthetic_alignment`: maximal coverage, weakest causal claim, and the flag says so. A user needing a strong claim filters to the first class; a user needing breadth accepts the third with its provenance explicit.

XII. DISCUSSION

What the composition buys an operator. It turns a network-security knob into a navigation-safety number. A fleet operator setting a detector’s false-positive budget currently reasons in packet statistics—“how much benign traffic can I afford to flag?”—with no way to connect that choice to airworthiness. Theorem 1 supplies the missing translation: each detector setting maps to a certified mission floor, so the operator can instead ask “what detector tightness do I need to certify my corridor?” and read the answer off Fig. 8.

Why the interface, not the theorem, is decisive. A reader might expect the certificate to be the hard part. Our results say otherwise: the tube is a short consequence of a classical inequality and the Lipschitz constant is measured, not assumed. What determines whether a useful detector setting is certifiable is γ , the metres-per-second of position error a second of staleness buys the attacker. The order-of-magnitude gap between the kinematic bound and the measured rate is the difference between a vacuous certified window and one that contains the operating point. This relocates the

research agenda: tightening cross-layer guarantees is primarily a *measurement* problem about how attacks propagate into state error—which is precisely why the benchmark and the theorem belong in one paper.

Two methodological findings we surface rather than bury. Building the benchmark forced two corrections that are instructive for anyone assembling a cross-layer corpus. First, our initial operating curve was contaminated by an accumulation bug in the sweep harness, in which each operating point inadvertently averaged all previous points; the tell was recall *rising* with the threshold in one regime, which is impossible for a threshold detector. The clean multi-seed campaign in Fig. 3 replaces it. Second, the bus dataset’s attack labels are not machine-readable per scenario, so rather than guess we derived them from frame statistics and flagged the derivation for confirmation. Both episodes argue *for* the provenance discipline the schema enforces: a benchmark that records how every number was obtained lets errors be caught rather than propagated.

Generality. Although we instantiate on a time-delay forwarding attack and a GNSS navigation model, the chain is agnostic to both ends. Any detector with a monotone operating point and any perturbation-to-state mapping slot into Lemma 1 and (5); the theorem and the four-certificate cover are unchanged. Extending the interface catalogue to jamming-intensity-to-error and to bus-level attacks is direct future work.

XIII. LIMITATIONS AND FUTURE WORK

We state limitations plainly; none undercuts the contributions, and each defines a concrete extension.

The delay-to-position rate is a three-flight hover calibration. The decisive quantity γ is measured on three real flights, all in a low-speed hover/loiter regime; at cruise speed the position error per second of staleness could be larger. This is the single result that could change the paper’s framing: if a cruise-inclusive calibration measured $\gamma > \gamma_{\text{req}} = 7.3 \text{ m/s}$ at a 10 m corridor—more than a five-fold increase over what we observe—the operating point would leave the certified window and the honest framing would revert to the narrow “certify-small, detect-large” regime that the conservative kinematic bound already guarantees. A larger, cruise-inclusive flight campaign is the highest-value follow-up.

Conservatism of the deterministic core. The Grönwall bound grows as e^{LT} , so for long horizons or higher Lipschitz constants the certified floor falls below what is empirically observed. This is by design—it is why the architecture is a four-certificate cover—but fusing the deterministic, smoothing, and PAC-Bayes floors into a single tighter envelope is open.

Evidence-class imbalance. Of the six datasets only one couples a network/RF event to a measured flight on one airframe, so it is the sole source of `measured_same_platform` pairings; the others support physics-modelled or synthetically-aligned couplings. This honestly reflects what public data exists, and is exactly why the honesty flag is mandatory rather than optional.

Calibration constants that remain data-gated. The unit bridge mapping a raw GNSS position error into receiver-level feature space (for the RF-manipulation classes, as opposed to the time-delay class the theorem targets) is calibrated on a synthetic signal corpus pending access to real in-phase and quadrature recordings; and the PAC-Bayes bound (11) is wired to the trained model’s measured empirical risk but awaits its numeric KL divergence. Neither gates the headline result; both are tracked explicitly so that no synthetic constant is reported as real.

Analytical navigation anchor and detector scope. The benchmark’s completion outcomes come from a physics-informed analytical Monte-Carlo backend calibrated to full-fidelity anchors; results should be described as a calibrated analytical benchmark, and we do so. We also instantiate the harness on one forwarding detector whose operating point is a single clean threshold, chosen because it makes the cross-layer sweep unambiguous; extending to multi-dimensional operating points requires a higher-dimensional sweep, not a schema change.

Threats to validity. *Construct:* does a recorded pairing represent the causal link it claims? The honesty flag bounds this risk, and a reviewer can restrict any analysis to the measured class. *Internal:* are the numbers correct? Two of our own measurement errors were caught precisely because the provenance discipline made them visible, and every figure is regenerated from committed data. *External:* four of six sources are simulation or calibrated twin, so absolute detector-performance numbers are corpus-specific; what transfers is the *structure*—the knee-and-floor operating curve and the order-of-magnitude gap between kinematic and empirical staleness—which is a property of store-and-forward scheduling and estimator behaviour, not of any one simulator.

XIV. ETHICS CONSIDERATIONS

This work involves no human subjects and no personally identifying data. All six datasets are public research corpora released by their authors for security research; we redistribute none of the raw third-party data, shipping only integration scripts and unified labels so that each source is fetched from its origin under its own licence. No live aircraft, production network, or third-party system was attacked: every attack in this paper is either replayed from a published capture or generated in simulation on infrastructure we control.

The dual-use question deserves a direct answer. Our operating-curve harness quantifies how an attacker can stay under a detector’s threshold, which is information an adversary could in principle use. We judge the disclosure clearly net-positive and consistent with standard practice in intrusion-detection research, for three reasons. First, the evasion strategy is not novel—staying under a published threshold is the definitional behaviour of an evading attacker, already assumed in the detector’s own analysis. Second, our central result *bounds* what such an attacker achieves and shows it saturates, which favours defenders. Third, the practical consequence we identify is a defensive configuration procedure: read the required detector

tightness off Fig. 8 for your corridor margin. We disclose no previously unknown vulnerability in a deployed system, so no coordinated-disclosure process was applicable; the underlying detector’s authors are aware of and have commented on this analysis of their operating point.

XV. OPEN SCIENCE AND ARTIFACT AVAILABILITY

Everything needed to reproduce this paper is released as an open artifact: the schema, the six ingestion adapters, the operating-curve harness, the certificate engine, the statistical analysis, and the figure generator. Every figure and table in this paper is machine-generated from committed result files rather than transcribed, so a reader can regenerate them from the released data; each carries a provenance tag (real-corpus, simulation-derived, or stated modelling parameter) in an artifact ledger. The harness is deterministic per seed, and headline comparisons are released per seed with their nonparametric tests. We intend to submit the artifact for evaluation. The repository is anonymised for review and will be de-anonymised on acceptance.

XVI. CONCLUSION

UAV security has been studied as two separate problems—attacks on the wire and attacks on the flight—because nothing let it be studied as one. We built the benchmark that makes the coupling representable and then proved, on exactly its objects, the guarantee that the coupling enables. The benchmark contributes a two-layer schema whose unit is the cross-layer pairing, six validated adapters, and a detector-operating-curve harness; the guarantee contributes a residual budget that measurement tightens into a *saturating* bound $\Delta(\theta) \leq H \min(\theta, s)$, an interface rate calibrated on real spoofing and jamming flights, and a composition theorem chaining a detector’s operating point to a certified mission-completion floor.

Three findings survive scrutiny. First, the composed guarantee is the only configuration that certifies a nonzero mission floor against a detector-evading attacker, dominating both single-layer baselines at every operating point with overwhelming statistical significance. Second, the empirically calibrated interface—an order of magnitude tighter than the kinematic worst case—brings the detector’s real operating point inside the certified window, so the guarantee is operationally relevant and not merely formal. Third, and most consequential for the field, the binding constraint on cross-layer certification is the measurement of how attacks propagate into state error, not the proof machinery: the interface, not the theorem, decides the outcome. UAV network security and UAV autonomy security, we conclude, are not two problems but two ends of one certificate.

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